

Database Technologies - Self-Learning

MySQL Lab Exercises

1. Explain EXISTS vs IN performance comparison with an example.
2. Implement a trigger to update the product price in cart, when price is updated in products table.

```
-- Users table
CREATE TABLE users (
  id INT PRIMARY KEY AUTO_INCREMENT,
  username VARCHAR(50) UNIQUE NOT NULL,
  email VARCHAR(100) UNIQUE NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Products table
CREATE TABLE products (
  id INT PRIMARY KEY AUTO_INCREMENT,
  name VARCHAR(255) NOT NULL,
  price DECIMAL(10, 2) NOT NULL,
  stock INT NOT NULL DEFAULT 0
);

-- Cart table
CREATE TABLE cart (
  id INT PRIMARY KEY AUTO_INCREMENT,
  user_id INT NOT NULL,
  product_id INT NOT NULL,
  price DECIMAL(10, 2) NOT NULL,
  quantity INT NOT NULL DEFAULT 1,
  FOREIGN KEY (user_id) REFERENCES users(id),
  FOREIGN KEY (product_id) REFERENCES products(id)
);
```

3. Understand ON DELETE CASCADE, ON UPDATE CASCADE. Write emp, dept CREATE TABLE statements and apply CASCADE for all foreign keys.

MongoDb Lab Exercises

1. Practice CRUD operations on Mongo.

1. Create two collections -- emp and dept.
2. Insert the emp and dept records.
3. Increase salary of dept 10 emps by 100.
4. Display emp name, job, deptno, sal in sorted order of deptno, job, sal (desc) for all emps whose sal more than 1000.
5. Delete the dept with id 40.

2. Learn grouping operations using Mongo aggregation pipeline

1. Display job wise total sal, max sal, min sal, and avg sal for all jobs other than PRESIDENT.
2. Display emp name, dept name, income (sal + comm) in sorted order of their income.

Leetcode Exercises

- GROUP BY Problems

1. 586. Customer Placing the Largest Number of Orders <https://leetcode.com/problems/customer-placing-the-largest-number-of-orders>
2. 619. Biggest Single Number <https://leetcode.com/problems/biggest-single-number>
3. 511. Game Play Analysis I <https://leetcode.com/problems/game-play-analysis-i>
4. 1693. Daily Leads and Partners <https://leetcode.com/problems/daily-leads-and-partners>
5. 1741. Find Total Time Spent by Each Employee <https://leetcode.com/problems/find-total-time-spent-by-each-employee>
6. 1050. Actors and Directors Who Cooperated at Least Three Times <https://leetcode.com/problems/actors-and-directors-who-cooperated-at-least-three-times>
7. 1890. The Latest Login in 2020 <https://leetcode.com/problems/the-latest-login-in-2020>
8. 1211. Queries Quality and Percentage <https://leetcode.com/problems/queries-quality-and-percentage>
9. 1141. User Activity for the Past 30 Days I <https://leetcode.com/problems/user-activity-for-the-past-30-days-i>
10. 596. Classes With At Least 5 Students <https://leetcode.com/problems/classes-with-at-least-5-students>

- JOINs Problems

1. Replace Employee ID With The Unique Identifier <https://leetcode.com/problems/replace-employee-id-with-the-unique-identifier>
2. Product Sales Analysis I <https://leetcode.com/problems/product-sales-analysis-i>
3. Employee Bonus <https://leetcode.com/problems/employee-bonus>
4. Students and Examinations <https://leetcode.com/problems/students-and-examinations>
5. Managers With At Least 5 Direct Reports <https://leetcode.com/problems/managers-with-at-least-5-direct-reports>
6. Last Person to Fit in the Bus <https://leetcode.com/problems/last-person-to-fit-in-the-bus>
7. Product Price at a Given Date <https://leetcode.com/problems/product-price-at-a-given-date>

- Subquery Problems

1. 181. Employees Earning More Than Their Managers <https://leetcode.com/problems/employees-earning-more-than-their-managers/>
2. 183. Customers Who Never Order <https://leetcode.com/problems/customers-who-never-order/>
3. 197. Rising Temperature <https://leetcode.com/problems/rising-temperature/>
4. 607. Sales Person <https://leetcode.com/problems/sales-person/>
5. 180. Consecutive Numbers <https://leetcode.com/problems/consecutive-numbers/>
6. 175. Combine Two Tables <https://leetcode.com/problems/combine-two-tables/>
7. 178. Rank Scores <https://leetcode.com/problems/rank-scores/>
8. 570. Managers with at Least 5 Direct Reports <https://leetcode.com/problems/managers-with-at-least-5-direct-reports>
9. 196. Delete Duplicate Emails <https://leetcode.com/problems/delete-duplicate-emails/>
10. 1204. Last Person to Fit in the Bus <https://leetcode.com/problems/last-person-to-fit-in-the-bus/>
11. 176. Second Highest Salary <https://leetcode.com/problems/second-highest-salary>

Written Exercises

1. Explain Conceptual, Logical and Physical data models for the tables: emp, dept, meeting, address.
2. Understanding locks: Shared locks (Read lock), Exclusive locks (Write lock), Row-level vs Table-level locks

CDAC Admission Process Case Study

- Design database for CDAC Admission System.
 - Create appropriate tables and insert sample records.
 - Write a Stored Procedure for Round1 allocation.
 - Write a Stored Procedure for Round2 allocation.
- Note:
 1. Create ER diagram following Normalization process.
 2. Study database constraints in depth: PRIMARY KEY, FOREIGN-KEY, UNIQUE, CHECK, DEFAULT
 - Apply appropriate constraints in CDAC admission system.
 3. Learn about composite keys and candidate keys
 - Identify keys in CDAC admission system.
 4. Explore AUTO_INCREMENT / SEQUENCES
 - Use appropriate AUTO_INCREMENT keys in CDAC admission system.
 5. Study control structures: IF-ELSE, CASE, LOOPS (WHILE, REPEAT, LOOP)
 - Use appropriate control structures in CDAC admission system stored procedures.
 6. Learn stored procedure parameters: IN, OUT, IN-OUT.
 - Use appropriate parameters for CDAC admission system stored procedures.
 7. Understand cursors and their usage in following stored procedures.
 - Make appropriate use of cursor in CDAC admission system stored procedures.
 8. Explore error handling in stored procedures
 - Implement appropriate error handling in CDAC admission system stored procedures.